

SIF Sentinel

AI-Assisted Serious Injury & Fatality Precursor Detection & Prevention Engine

TECHNICAL BRIEFING &
ARCHITECTURE DOSSIER

Executive Summary: SIF Sentinel is an enterprise safety decision-support platform designed for high-risk industrial environments (mining, energy, manufacturing, construction). It detects hidden precursors to Serious Injuries and Fatalities (SIF) from incident descriptions, safety observations, and near-miss logs before catastrophe strikes.

1. WHAT SIF SENTINEL SOLVES

Traditional safety management systems suffer from the "**Heinrich's Pyramid Fallacy**"—the outdated assumption that reducing minor trivial incidents (e.g. slips, paper cuts) proportionally reduces fatal accidents. Modern safety science proves that **only ~20% of near-misses contain the precursor conditions that lead to fatalities** (e.g., heavy machinery interaction, high-pressure releases, suspended loads, confined space entry, electrical lockout failures).

SIF Sentinel replaces slow manual triage with a **hybrid Multi-Factor Decision Engine**:

- **Dual-Layer NLP Classification:** Extracts domain hazards (Electrical, Height, Heavy Equipment, Chemical), life-critical activities, and barrier health indicators from natural language narratives.
- **Deterministic Safety Rules Engine:** Enforces non-negotiable safety rules (e.g., Lockout/Tagout, Confined Space permits, Fall Arrest compliance). Even if an ML probability is low, rule violations immediately escalate the incident to High Priority.
- **Composite Risk Scoring & Prioritization:** Computes an explainable composite risk score (\$0.00 - 1.00\$) categorizing reports into HIGH, MEDIUM, or LOW priority for immediate HSE officer triage.
- **Human-in-the-Loop (HITL) Auditability:** Provides side-by-side comparison between AI suggestions and safety officer verdicts, capturing feedback in an immutable, append-only audit trail.
- **Vector Semantic Similarity Search:** Automatically surfaces historical near-misses and recurring hazards using vector cosine embeddings (Sentence Transformers).

2. SYSTEM ARCHITECTURE & TECH STACK

BACKEND & DECISION ENGINE

FastAPI + PyTorch

Uvicorn ASGI, SQLAlchemy 2.0 ORM, SQLite/PostgreSQL, Pydantic v2 schemas, HuggingFace Transformers.

FRONTEND & USER EXPERIENCE

React 18 + Vite

Tailwind CSS design system, Lucide icons, responsive dashboard with real-time triage queue and audit timeline.

Subsystem	Technology	Role & Key Capabilities
Machine Learning Core	Fine-Tuned DistilBERT	Evaluates text narratives to predict SIF precursor likelihood with high recall ($F_2 = 0.9615$, 100% recall).
Semantic Memory	Sentence-Transformers (MiniLM)	Generates 384-dimensional vector embeddings for near-miss semantic matching and pattern detection.
Rules Engine	Deterministic Python Engine	Evaluates 10 critical industrial life-saving rules; ensures zero AI hallucinations bypass safety policy.
Storage & Audit	SQLAlchemy 2.0 / SQLite	ACID transactions, relational schema, append-only audit ledger tracking every inference and HSE modification.
REST API Layer	FastAPI / OpenAPI 3.1	23 registered endpoints for batch analysis, real-time ingestion, HSE reviews, audit trails, and KPI telemetry.

3. MACHINE LEARNING PERFORMANCE & VALIDATION RESULTS

During rigorous automated validation across all 508 tests in the test suite, SIF Sentinel demonstrated complete stability and zero regressions:

TOTAL AUTOMATED TESTS

508 / 508 PASSED

281 Backend, 145 ML, 60 Dataset, 22 Frontend (100% Pass Rate).

PRECURSOR DETECTION RECALL

100.0% (\$F_2 = 0.9615\$)

Zero false negatives on independent held-out test split.

Independent Test Split Metric Comparison

Evaluation Metric	Baseline (TF-IDF + LogReg)	SIF Sentinel (Fine-Tuned DistilBERT)	Operational Significance
\$F_2\$ Score (Primary)	0.9259	0.9615 (+0.0356)	Weights recall 2x over precision—missed SIF precursors are catastrophic.
Recall (\$TPR\$)	100.0%	100.0%	Zero True SIF precursors missed (0 False Negatives).
Precision	71.43%	83.33% (+11.9%)	Reduces false alerts by 50%, preventing alert fatigue among HSE officers.
ROC-AUC	0.8000	0.8667 (+0.0667)	Superior confidence ranking across probability thresholds.

4. CORE USER WORKFLOWS & KEY FEATURES

- **Live Incident Triage Queue:** Sorts incident feeds dynamically by Composite Priority Score, surfacing imminent danger observations first.
- **Multi-Factor Explainability:** Breaks down why an incident was flagged (e.g. "Lockout/Tagout Rule Triggered: Breaker energization without padlock" + "ML High SIF Probability (89%)").
- **Human-in-the-Loop Override:** Allows certified HSE Officers to confirm, override, or re-classify risk levels with required justification notes stored permanently in the audit ledger.
- **Semantic Near-Miss Matching:** One-click lookup displays previously recorded incidents with similar text patterns, helping investigators detect systemic equipment or contractor failures.
- **Executive KPI Analytics:** Aggregates high-potential rate, active review backlogs, and recurring hazard distributions across sites and shifts.

5. HOW TO RUN SIF SENTINEL LOCALLY (WINDOWS 11)

```
# 1. Start the Backend API Server (Terminal 1) cd c:\Users\ronak\OneDrive\Desktop\SIF-Sentinel\backend
.\venv\Scripts\Activate.ps1 uvicorn app.main:app --reload --host 127.0.0.1 --port 8000 # 2. Start the
Frontend Application (Terminal 2) cd c:\Users\ronak\OneDrive\Desktop\SIF-Sentinel\frontend npm run dev #
```

```
3. Access in Web Browser: # Frontend UI: http://127.0.0.1:5173 # Backend Docs:  
http://127.0.0.1:8000/docs
```